Pengembangan Temporal Graph Neural Network untuk Deteksi Gegenpressing pada Data Spatiotemporal IDSSE

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Abstrak

Deteksi gegenpressing memerlukan pengamatan respons kolektif pemain segera setelah kehilangan penguasaan bola. Penelitian ini mengembangkan pipeline deteksi berbasis Temporal Graph Neural Network pada Integrated Dataset of Spatiotemporal and Event Data in Elite Soccer (IDSSE) serta menyiapkan logistic regression dan XGBoost sebagai pembanding. Pendekatan yang digunakan adalah penelitian kuantitatif dengan eksperimen komputasional komparatif dan pembelajaran terawasi. Data tujuh pertandingan diolah menjadi kandidat transisi, disaring berdasarkan kelengkapan tracking, kemudian direpresentasikan sebagai urutan graph pemain dan bola. Setiap episode mencakup dua detik sebelum hingga dua detik sesudah kehilangan bola pada frekuensi 25 Hz, menghasilkan 101 frame dengan 23 node dan enam fitur per node. Arsitektur model menggabungkan dua lapisan Graph Convolutional Network, pooling per frame, dan Gated Recurrent Unit. Pemeriksaan hingga 19 Oktober 2026 menghasilkan 2.353 kandidat dan 1.873 episode valid ketat. Dari 264 episode dalam antrean anotasi, 132 telah diisi, terdiri atas 105 gegenpressing, 16 bukan gegenpressing, dan 11 ragu; 121 label biner tersedia. Pipeline telah menjalani uji integrasi pelatihan dan evaluasi, tetapi hasil tersebut berasal dari 26 label pilot. Anotasi validation dan test masing-masing baru mencakup tiga episode sehingga evaluasi akhir belum layak dilakukan. Capaian saat ini mendukung kelayakan teknis pembentukan dataset dan model. Keunggulan Temporal GNN masih memerlukan penyelesaian anotasi, audit label, penyelarasan representasi spasial, dan eksperimen akhir berbasis pemisahan pertandingan.

Kata kunci: gegenpressing; graph neural network; data tracking; IDSSE; anotasi manual

Abstract

Gegenpressing detection requires observing collective player responses immediately after possession loss. This study develops a temporal graph neural network pipeline using the Integrated Dataset of Spatiotemporal and Event Data in Elite Soccer (IDSSE), with logistic regression and XGBoost as comparative models. The study adopts a quantitative, comparative computational design with supervised learning. Tracking data from seven matches are converted into transition candidates, screened for structural completeness, and represented as player–ball graph sequences. Each episode spans two seconds before and two seconds after possession loss at 25 Hz, yielding 101 frames, 23 nodes, and six node features. The architecture combines two graph convolutional layers, frame-level pooling, and a gated recurrent unit. As of 19 October 2026, the pipeline identified 2,353 candidates and 1,873 strictly valid episodes. Of 264 selected episodes, 132 have been annotated: 105 positive, 16 negative, and 11 uncertain, leaving 121 binary labels. Training and evaluation integration tests were completed using 26 pilot labels. However, the current validation and test sets each contain only three annotated episodes, preventing a meaningful final evaluation. These findings establish the technical feasibility of dataset preparation and model

execution. Any claim of improved detection performance requires complete annotation, label auditing, spatial-representation checks, and final match-level evaluation.

Keywords: gegenpressing; graph neural network; tracking data; IDSSE; manual annotation

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